

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0007

Belleair 138 kV
April 22, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0007. The Interconnection Customer proposes a 63.7 MW battery energy storage facility interconnecting at the Belleair 138 kV substation in the Virginia area. The request was received on January 25, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 4 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$141,022,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ROSS FORK - ARTESIA COUNTRY CLUB 138 kV	ARTESIA COUNTRY CLUB - AIR FORCE BASE 345 kV	121.3	231	6.5
ROBISON PARK - RUSSEL FORK 230 kV	DEKALB SUBSTATION - JONES CHAPEL 500 kV	120.5	557	7.8
CONCORD - BLUEBONNET 230 kV	DEKALB - JONES CHAPEL 500 kV	118.5	499	5.2
BLUEBONNET - JONES CHAPEL 230 kV	ROBISON PARK - RUSSELL FORK 230 kV	116.7	575	5.1
LORETTO - CROCKETT 138 kV	URBAN - AMERICAN 345 kV	116.3	380	7.9
GRAVEL RIDGE - SHEPHERDSTOWN 138 kV L	AMERICAN - URBAN 345 kV line	115.1	298	7.3
DAM NO. 4 HYDRO STA - GRVEL RIDGE 138 kV L	AMERICAN - URBAN 345 kV	114.8	304	7.3
MIDSET - EDENS RIDGE 138 kV	AMERICAN - URBAN 345 kV	114.6	190	7.0
WHEELER - PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 kV	LOUDON - THOMASTON SUBSTATION 345 kV	110.9	231	11.2
BLUEBONNET - RUSSELL FORK 230 kV	DEKALB - JONES CHAPEL 500 kV	108.6	486	5.7
WEST WINCHESTER - INTERNATIONAL PAPER VALLIANT OK 138 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345 kV	108.4	307	10.8
MIDSET - BROOKLYN CITY NORTH PLANT 138 kV	AMERICAN - URBAN 345 kV	104.2	186	6.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TWIN LAKES ALTE - T.S. POWER 138 kV	THOMASTON - LOUDON 345 kV line	101.0	185	16.7
EDENS RIDGE - FIREWHEEL SUBSTATION 138 kV	AMERICAN - URBAN 345 kV	100.8	242	7.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ROSS FORK - ARTESIA COUNTRY CLUB 138 kV	Rebuild line with higher-capacity conductor	\$3,609,000	100% (\$3,609,000)
ROBISON PARK - RUSSEL FORK 230 kV	Rebuild line with higher-capacity conductor	\$41,441,000	100% (\$41,441,000)
CONCORD - BLUEBONNET 230 kV	Rebuild line with higher-capacity conductor	\$22,384,000	100% (\$22,384,000)
BLUEBONNET - JONES CHAPEL 230 kV	Rebuild line with higher-capacity conductor	\$24,051,000	100% (\$24,051,000)
LORETTO - CROCKETT 138 kV	Rebuild line with higher-capacity conductor	\$4,596,000	100% (\$4,596,000)
GRAVEL RIDGE - SHEPHERDSTOWN 138 kV L	Rebuild line with higher-capacity conductor	\$1,766,000	100% (\$1,766,000)
DAM NO. 4 HYDRO STATION - GRAVEL RIDGE 138 kV	Rebuild line with higher-capacity conductor	\$3,738,000	100% (\$3,738,000)
MIDSET - EDENS RIDGE 138 kV	Rebuild line with higher-capacity conductor	\$11,053,000	100% (\$11,053,000)
WHEELER - PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 kV	Rebuild line with higher-capacity conductor	\$4,068,000	100% (\$4,068,000)
RUSSELL FORK - BLUEBONNET SUBSTATION 230 kV line	Reconductor line	\$15,605,000	100% (\$15,605,000)
WEST WINCHESTER - INTERNATIONAL PAPER VALLIANT OK 138 kV	Reconductor line	\$679,000	100% (\$679,000)
MIDSET - BROOKLYN CITY NORTH PLANT 138 kV	Reconductor line	\$3,989,000	100% (\$3,989,000)
T.S. POWER - TWIN LAKES ALTE 138 kV	Reconductor line	\$1,251,000	100% (\$1,251,000)
EDENS RIDGE - FIREWHEEL 138 kV L	Reconductor line	\$2,792,000	100% (\$2,792,000)
	Total allocated to project		\$141,022,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.